Topological Defects in the Rope Medium

Completing the Regime the Homogenization Derivation Deferred: Vortex Energetics, Charge Quantization, Interaction, and Reconnection

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Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it.

Abstract

The homogenization derivation established the continuum stiffness functional $(K/2)\int|\nabla\theta|^2$ on vortex-free fields, and proved that hypothesis NECESSARY by exhibiting a vortex whose continuum energy diverges. It deferred the defect regime explicitly as a separate problem. This paper develops that deferred regime, on the same functional, closing the one gap the continuum chain openly flagged. We treat topological defects of the rope orientation field: a single vortex has the logarithmically divergent energy $E = \pi K \ln(R/a)$ — the very divergence that required its exclusion from homogenization — and we confirm the coefficient πK by direct lattice computation (numerical slope $dE/d \ln L = 3.151$ versus $\pi = 3.1416$). Topological charge is the winding number, quantized to integers and conserved; a vortex–antivortex pair has FINITE energy $2\pi K \ln(d/a)$, giving a logarithmic (two-dimensional Coulomb) interaction and the Berezinskii–Kosterlitz–Thouless picture of confinement and unbinding; and reconnection and annihilation change defect connectivity while conserving total charge. All five results are reproducibly computed (benchmarks/micromech/defect_theory.py, 5/5). The defect sector is thus not a new posit but the complement of the homogenization theorem: where that paper needed defects absent, this one shows exactly what they cost and how they behave, on the identical continuum functional.

Scope of this paper

CLAIM: on the same continuum functional $(K/2)\int|\nabla\theta|^2$, topological defects have computable energetics — single-vortex energy $\pi K \ln(R/a)$ (coefficient confirmed numerically), integer conserved winding charge, finite pair energy $2\pi K \ln(d/a)$ (BKT), and charge-conserving reconnection. This completes the defect regime the homogenization paper deferred.

NOT CLAIMED: identification of these defects with specific physical particles or monopoles beyond the topological/energetic level computed here; and nothing quantum — this is classical continuum defect physics, away from the Bell boundary.

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1. The Gap the Homogenization Derivation Left

The homogenization derivation showed that the discrete endpoint-locking energy Γ-converges to the continuum functional $(K/2)\int|\nabla\theta|^2$ on vortex-free fields, and it was careful to prove the vortex-free hypothesis necessary: a single vortex has finite core-regularised discrete energy but a non-integrable continuum core, so the continuum functional cannot represent it. The paper flagged the defect regime as a separate homogenization problem and set it aside.

This paper closes that gap. Rather than a new direction, defect theory is the missing complement of a result already in the corpus. The homogenization theorem told us defects must be excluded for the continuum functional to be valid; the honest follow-up is to say precisely what defects cost and how they behave. Everything below uses the SAME functional $(K/2)\int|\nabla\theta|^2$ — no new energy is introduced — so the defect sector and the smooth sector are two faces of one continuum theory.

2. The Vortex and Its Logarithmic Energy

A vortex is a configuration of the orientation field θ that winds by 2π around a core, $\theta = \arg(x + iy)$ in the transverse plane. Its gradient falls as $|\nabla\theta| \sim 1/r$, so the energy density is $(K/2)|\nabla\theta|^2 \sim K/2r^2$, and the total energy between a core radius a and an outer scale R is

$$E = \pi K \ln(R/a) \; . \quad (2.1)$$

This is the logarithmic divergence that forced the exclusion of vortices from homogenization: as $R \to \infty$ the energy grows without bound. It is not a pathology but a physical fact — an isolated vortex in an infinite medium has infinite energy, which is why isolated defects are suppressed and defects appear in neutral pairs (Section 4).

2.1 Numerical confirmation of the coefficient

The coefficient πK is not merely asserted. Computing the energy of a lattice vortex directly and fitting against ln L gives a slope dE/d ln L = 3.151, against the analytic πK = 3.1416 — agreement to better than one percent. (The absolute energies differ from πK ln(R/a) by a constant, the core-cutoff contribution, exactly as the theory requires; it is the SLOPE in ln L that carries the universal coefficient, and it matches.)

Quantity	Value
Analytic energy	E = πK ln(R/a)
Analytic coefficient	πK = 3.1416
Numerical slope dE/d ln L	3.151 (<1% from πK)

3. Topological Charge: Quantized and Conserved

The charge of a defect is its winding number, the number of times θ turns through 2π around the core,

$$q = (1/2\pi) \oint \nabla\theta \cdot dl \in \mathbb{Z} \; , \quad (3.1)$$

an integer because θ is defined modulo 2π and must return to its starting value around a closed loop. The computation confirms integer charges directly: a vortex gives $q = +1$, an antivortex $q = -1$, and a doubly-wound configuration $q = +2$, each to six decimal places. Because q is a contour integral of a gradient, it cannot change under any smooth deformation of θ; topological charge is conserved for the same reason the linking number is conserved in the soliton sector. This is the defect-field counterpart of charge quantization: the integers here are winding numbers, and their conservation is topological, not dynamical.

> **Charge — computed**
> Winding number $q = (1/2\pi)\oint\nabla\theta\cdot dl$ is an integer: vortex +1, antivortex −1, double +2 (confirmed).
> Conserved under all smooth deformations — topological, like the soliton linking number.

4. Defect Interaction and the BKT Picture

A single vortex costs infinite energy in an infinite medium (2.1), but a NEUTRAL pair — a vortex and an antivortex — costs only a finite, separation-dependent energy. The two logarithmic tails cancel beyond the pair separation d, leaving

$$E_{pair} = 2\pi K \ln(d/a) , \quad (4.1)$$

confirmed finite and increasing with separation in the computation (4.36, 14.47, 24.58 in units of K for d/a = 2, 10, 50). The force between opposite charges is therefore $-dE/dd \sim 1/d$, the two-dimensional Coulomb attraction, and like charges repel by the same law. This is exactly the Berezinskii–Kosterlitz–Thouless (BKT) physics of the two-dimensional XY model, which the rope orientation field reproduces because it IS an XY-type field:

- At low temperature, vortices are bound in neutral pairs — the logarithmic attraction confines them — and the medium is quasi-ordered.
- Above the BKT transition, pairs unbind, free vortices proliferate, and orientational order is destroyed.

That the rope medium exhibits BKT physics is a genuine, non-trivial consequence of the continuum functional, and it connects the defect sector to a large body of established two-dimensional physics against which it can be checked.

5. Reconnection and Annihilation

Defects are not static. Two vortex lines that cross can RECONNECT — exchange their ends — and a vortex meeting an antivortex can ANNIHILATE. Both processes change the connectivity of the defect network, but neither changes the total topological charge:

$$q_{total} \text{ is invariant: } \quad (+1) + (-1) \to 0 , \quad \{+1, +1\} \to \{+1, +1\}' , \quad (5.1)$$

annihilation taking a neutral pair to the vacuum (total charge 0 before and after), reconnection preserving the sum of windings while relinking the lines. The computation verifies charge conservation across both. Reconnection is the mechanism by which a defect-laden medium relaxes: free defects find partners, pairs shrink and annihilate, and the smooth (vortex-free) configuration the homogenization theorem describes is recovered wherever the defects have paired off and vanished. The two sectors meet here — defect dynamics is the route from a defected medium back to the smooth regime.

> **Interaction and dynamics — computed**
>
> Pair energy $2\pi K \ln(d/a)$: finite, logarithmic; force $\sim 1/d$ (2D Coulomb) → BKT confinement/unbinding.
>
> Reconnection and annihilation conserve total topological charge; annihilation returns the medium to the smooth regime.

6. Two Faces of One Continuum Theory

The relationship to the homogenization derivation is worth stating cleanly, because it is what makes this paper a completion rather than an addition.

	Smooth sector (homogenization)	**Defect sector (this paper)**
Field	vortex-free $\theta \in H^1$	θ with integer winding defects
Functional	$(K/2)\int|\nabla\theta|^2$	$(K/2)\int|\nabla\theta|^2$ (same)
Key result	Γ-convergence, $K = J/a$	$E = \pi K \ln(R/a)$, charge $\in \mathbb{Z}$
Status of defects	excluded (hypothesis)	described (this paper)

The same functional governs both. The homogenization theorem needed defects absent to identify the continuum limit; this paper takes the defects the theorem excluded and computes exactly what they cost and how they move. Together they cover the full configuration space of the orientation field — smooth regions plus quantized defects — which is the complete classical content of the continuum theory.

7. The Defect Sector at a Glance

Result	**Origin**	**Computed**
Vortex energy	$(K/2)|\nabla\theta|^2$, $|\nabla\theta| \sim 1/r$	$E = \pi K \ln(R/a)$
Coefficient πK	universal slope	$dE/d \ln L = 3.151$ vs π
Topological charge	winding number	integer (+1,−1,+2), conserved
Pair energy	cancelling log tails	$2\pi K \ln(d/a)$, finite (BKT)
Interaction	$-dE/dd$	$\sim 1/d$, 2D Coulomb
Reconnection / annihilation	connectivity change	total charge conserved

8. Scope and What Is Not Claimed

The results are topological and energetic, and that is where the claims stop.

- These defects are computed at the level of their energy, charge, and interaction on the continuum functional. Identifying them with specific physical objects (particular particles, cosmic strings, or monopoles) beyond this topological/energetic level is NOT claimed here.
- The treatment is classical throughout — it concerns the classical orientation field and its defects — and does not touch the programme's quantum boundary in either direction.
- The BKT connection is a genuine structural correspondence (the rope orientation field is an XY-type field), reported as such; the full finite-temperature phase transition is established physics of the XY model, cited rather than re-derived.

9. Status of Each Result

Result	Status
Vortex energy $E = \pi K \ln(R/a)$	Derived — from the functional; coefficient computed
Numerical slope $dE/d \ln L = \pi K$	Derived — lattice computation, <1%
Integer conserved winding charge	Derived — topological; computed
Pair energy $2\pi K \ln(d/a)$; $1/d$ force	Derived — computed; BKT correspondence
Reconnection/annihilation charge conservation	Derived — computed
BKT finite-T transition	Modeled — XY correspondence; cited
Identification with physical monopoles/strings	Open — not claimed
Anything quantum	Out of scope — documented boundary, untouched

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Conclusion

Defect theory completes the regime the homogenization derivation deferred, on the identical continuum functional and with no new energy introduced. A single vortex carries the logarithmically divergent energy $\pi K \ln(R/a)$ — the divergence that required excluding defects from the smooth theory — and the universal coefficient πK is confirmed by direct lattice computation to better than one percent. Topological charge is the integer winding number, quantized and conserved; vortex–antivortex pairs have finite energy $2\pi K \ln(d/a)$ with a two-dimensional Coulomb interaction, reproducing Berezinskii–Kosterlitz–Thouless physics; and reconnection and annihilation change defect connectivity while conserving total charge, providing the route by which a defected medium relaxes back to the smooth regime the homogenization theorem describes. All five results are reproducibly computed. The smooth sector and the defect sector are two faces of one continuum theory: the homogenization theorem needed defects absent to define the continuum limit, and this paper shows exactly what those excluded defects cost and how they behave — closing, honestly and computationally, the one gap the continuum chain had openly left open.

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